REPLASTIX INNOVATIONS: A SALESFORCE-DRIVEN SUSTAINABLE WASTE MANAGEMENT SOLUTION

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1. Understand the Goal

You need to build a **Salesforce Waste Management System** for RePlastix Innovations that:

- Automates plastic waste collection and processing
- Tracks recycled materials in real-time
- Improves customer and partner engagement
- Ensures environmental compliance and data security
- Provides reports & sustainability analytics

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2. Design the Data Model (VERY IMPORTANT)

Start with objects (tables in Salesforce):

Core Objects

- Customer (Account / Contact)
- Waste Collection
- Recycling Order
- Recycling Line Item
- Inventory (Recycled Materials)
- Supplier / Vendor (optional)

Relationships

- One Customer → Many Waste Collections
- One Collection → Many Recycling Orders
- One Order → Many Line Items
- One Material → Many Inventory Records

Example:

Recycling Order

- Order ID

- Status
- Total Weight
- Processing Cost

Recycling Line Item

- Waste Type
- Quantity
- Processing Method

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3. Create Fields

Recycling Order Object

- Order Status (Picklist: Collected, Processing, Recycled, Delivered)
- Collection Date
- Total Weight (Formula)

Inventory Object

- Material Available
- Reuse Level
- Last Updated

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4. Formula Fields

Use formulas for automation:

- **Total Weight**
  Quantity * Material Weight
- **Recycling Efficiency**
  IF(Material_Available < Reuse_Level, "Low Efficiency", "Optimal")

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5. Validation Rules

Prevent incorrect data entry:

- Quantity <= 0 → Show Error:
  "Waste quantity must be greater than zero"

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6. Automation (Flow / Process Builder)

You MUST include automation:

When Waste is Collected

- Create processing task
- Send notification to recycling unit

When Status = "Processing"

- Allocate recycling resources automatically

When Material is Low

- Send alert to admin / supplier

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7. Apex (IMPORTANT for your project)

Use Apex for complex automation.

Apex Trigger Example

- Trigger on Recycling Order → update inventory

Pseudo Logic:

```
for(Order o : Trigger.new){
  for(LineItem li : o.LineItems){
    Inventory__c inv = getInventory(li.MaterialId);
    inv.Stock__c += li.Quantity;
    update inv;
  }
}
```

Apex Class

- Handles recycling calculations, notifications

Test Class

- Minimum 75% code coverage required

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8. Security (VERY IMPORTANT)

Implement:

Profiles & Roles

- Admin
- Field Worker
- Recycling Manager

Field-Level Security

- Restrict sensitive environmental data

Sharing Rules

- Only assigned teams can access records

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9. User Management

- Create Users
- Assign Roles
- Assign Profiles

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10. Reports & Dashboards

Reports

- Waste Collected by Type
- Recycling Efficiency
- Inventory Levels

Dashboard

- Total Waste Processed
- Active Recycling Orders
- Sustainability Metrics

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11. Integration (Basic Level)

Simulate:

- Email notifications
- Customer complaint case creation

- IoT-based waste tracking (optional idea)

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12. Final Use Case Flow (Explain in Submission)

Customer disposes plastic waste
→ Waste collection request created
→ Collection team assigned automatically
→ Waste processed and recycled
→ Inventory updated
→ Notifications sent
→ Reports & sustainability data updated

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13. What You Should Submit

Your project should include:

- Data Model Diagram
- Objects & Fields List
- Automation (Flow / Process Builder)
- Apex Code + Test Class
- Security Setup
- Sample Reports & Dashboard
- Use Case Explanation